University of Toronto

Faculty of Applied Science & Engineering

MIE353

Assignment 2 Deliverable - Written Questions 1 and 2

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1.0 Exploring The Course Management System (CMS)

In this section I will answer the questions from Part 1 of the Assignment to explore and better understand the starter code.

1.1 How many entity classes are there in the schema (org.example.schema.CMS.)?*

There are 5 entity classes, which are Course, Professor, Student, Teaches, Transcript. I found this by inspecting the cms-ddl.sql file in the GitHub starter code, which contains five CREATE TABLE definitions, to make the five respective entity classes under CMS.

1.2 What purpose does the cms-ddl.sql file serve? Where is it used?

The purpose of the cms-ddl.sql file is to define the database schema with the tables, columns, primary keys, and foreign-key relations. When looking at where the cms-ddl.sql file is used with the "search this repository" feature, CMS.dbconfig references it through a dbDdl property, to create the structure of the database objects in the database [1].

1.3 What are the foreign keys of the transcript table?

When looking at the transcript table found in cms-ddl.sql, it uses two foreign keys which are studentId referencing student(studentId) and courseId referencing course(courseId).

1.4 How many parameters does the FindStudent query have?

The FindStudent query has only one parameter which is the parameter studentId based on the code in FindStudent.sql and Main.java.

1.5 What format is the database stored in?

When observing the CMS.dbconfig file, the url is defined as jdbc:duckdb: and inMemory is set to true which defines the database to be an in-memory DuckDB relational database system [2]. Additionally, its initial data is stored in CSV files and then loaded into the memory.

2.0 Fixing Software Problems

Refer to the changes made in the fischmei-MIE353-assignment2-2026-starter.

3.0 References

[1] GeeksforGeeks, “SQL Commands | DDL, DQL, DML, DCL and TCL Commands,” GeeksforGeeks, Nov. 06, 2017. https://www.geeksforgeeks.org/sql/sql-ddl-dql-dml-dcl-tcl-commands/ (accessed Nov. 04, 2026).

[2] “Java (JDBC) Client,” DuckDB. https://duckdb.org/docs/2.0/clients/java/overview.html (accessed Nov. 03, 2026).